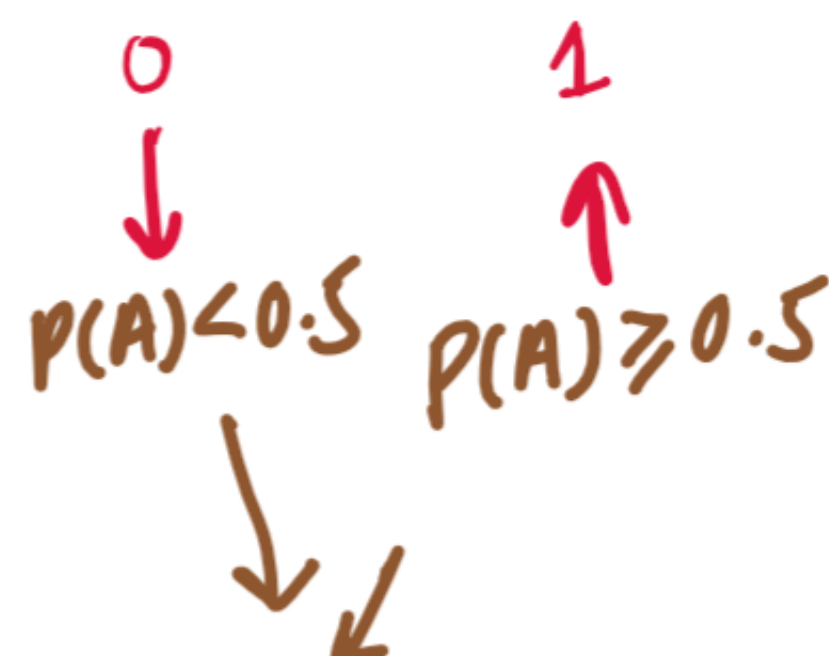


LOGISTIC REGRESSION

- CLASSIFICATION ALGO. ($\hat{y} \rightarrow \text{Binary}(0 \text{ or } 1)$)
- Based on the probability of a sample to be a class.
- Probabilities must be continuous and bounded between (0,1)



SLR $\hat{y} = \beta_0 + \beta_1 x_1 + \epsilon$

MLR $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$

NUMERIC / CONTINUOUS

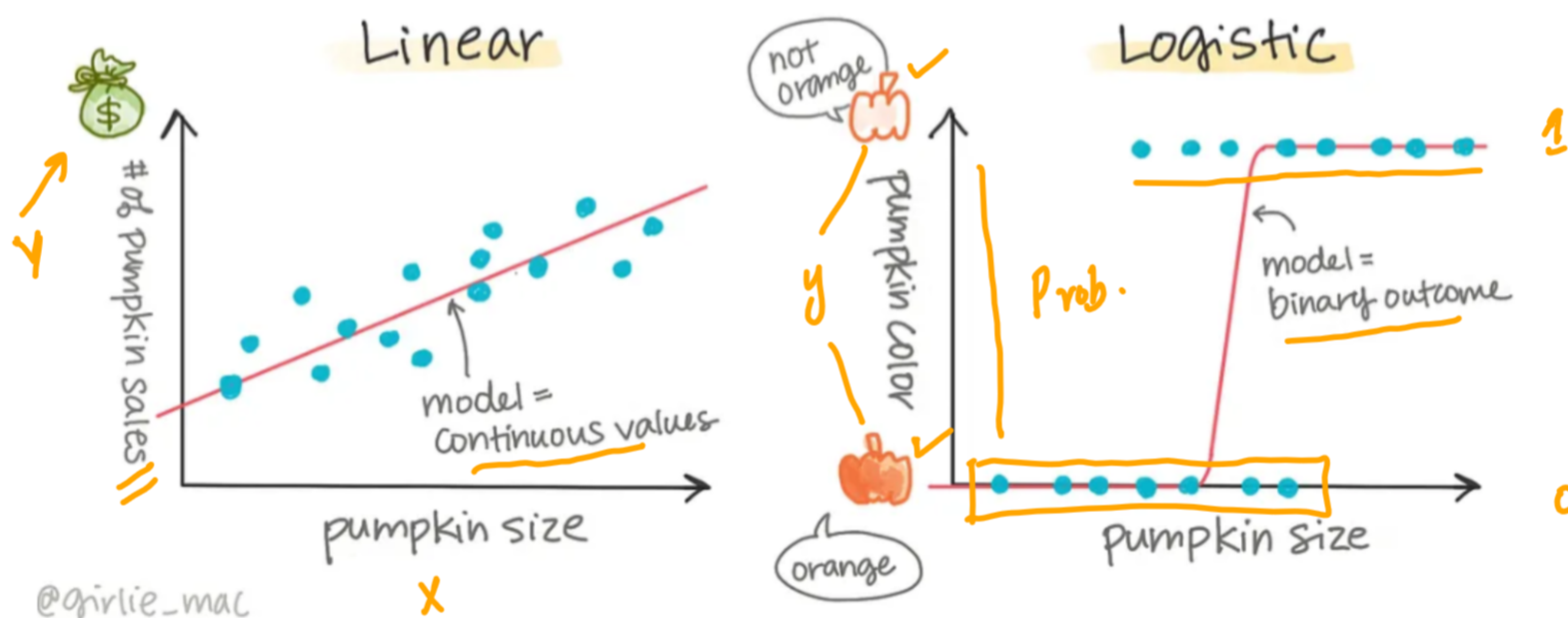
• Prob. of an Event
 $P(A) = \frac{n(A)}{n(S)} ; 0 \leq P(A) \leq 1$

IMPORTANT CONCEPTS TO UNDERSTAND LOGISTIC REG.

- ODD RATIO $\rightarrow OR = \frac{P}{1-P} = \frac{P(\text{occurrence})}{P(\text{Non-occurrence})}$
- LOGIT FUNCTION $\rightarrow \log(OR) \Rightarrow \text{logit}(P) = \log\left(\frac{P}{1-P}\right)$
- SIGMOID FUNCTION $\rightarrow f(x) = \sigma_x = \frac{1}{1+e^{-x}}$

log(??)

LINEAR vs. LOGISTIC REGRESSION



ODD RATIO:

Odds Ratio (OR) is the odds in favor of a particular event. It is a measure of association between exposure and outcome.

$$OR = \frac{P}{1-P} = \frac{x/(x+y)}{y/(x+y)}$$

Exp: Suppose that 7 out of 10 boys are admitted to Data Science while 3 of 10 girls are admitted. Find the odds of boys getting admitted to Data Science?

$n(\text{Boys}) = 7$ $n(\text{Girls}) = 3$ $n(S) = \text{Total} = 10$

$P(\text{Boys}) = \frac{n(\text{Boys})}{n(S)} = \frac{7}{10}$ $P(\text{Boys}') = 1 - P(\text{Boys}) = 1 - \frac{7}{10} = \frac{3}{10}$

$P(\text{Girls}) = \frac{n(\text{Girls})}{n(S)} = \frac{3}{10}$ $P(\text{Girls}') = \frac{7}{10}$

$\text{odds}(\text{Boys}) = \frac{P(\text{Boys})}{P(\text{Boys}')} = \frac{7/10}{3/10} = \frac{7}{3} = 2.33$

$\text{odds}(\text{Girls}) = \frac{P(\text{Girls})}{P(\text{Girls}')} = \frac{3/10}{7/10} = \frac{3}{7} = 0.42$

$\text{odds}(\text{boys admission}) = \frac{\text{odd}(\text{boys})}{\text{odd}(\text{female})} = \frac{2.33}{0.42} = 5.54$

$\text{odd}(\text{female admission}) = 0.42 / 2.33 = 0.18$

LOGIT FUNCTION:

- Let consider that a house is sold or not on the base of price?
- Regression equation for this use case will be like

The output must be $(0, 1)$ "not sold" "sold"

$$\hat{y} = \beta_0 + \beta_1(\text{price}) + \epsilon$$

So this also takes the prob. form as of odds $\frac{P}{1-P}$

Logit function will takes it to the range of $(0, 1)$

$$\log(\hat{y}) = \beta_0 + \beta_1(\text{price}) + \epsilon_i$$

hence $\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1(\text{price}) + \epsilon_i$

ADDITION

$$\log(5.44)$$

$$\downarrow$$

$$0 - 1$$

EXAMPLE:

Suppose there is a car and its price is \$60,000 and it has an extra feature of 1 air bag

Find the probability of selling the car?

price airbags

Model

6 → rationalize value e.g. if it is 90,000 it will rationalize as 9

Sell
y → NO

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Suppose our model has following outlook:

$$\hat{y} = 0.231 + 0.5(x_1) + 0.9(x_2)$$

S-I. OR

$$\frac{P}{1-P} = 0.231 + 0.5(x_1) + 0.9(x_2)$$

S-II. logit func.

$$\log\left(\frac{P}{1-P}\right) = 0.231 + 0.5(x_1) + 0.9(x_2)$$

$$= 0.231 + 0.5(6) + 0.9(1)$$

$$\log\left(\frac{P}{1-P}\right) = 4.131$$

$$\frac{P}{1-P} = \log^{-1}(4.131)$$

$$\frac{P}{1-P} = e^{4.131} = 5.677$$

Probability

$$\rightarrow P = 5.677(1-P)$$

$$P = 5.677 - 5.677P$$

$$P + 5.677P = 5.677 \Rightarrow 6.677P = 5.677$$

$$P = 5.677/6.677$$

$P = 0.85 \rightarrow$ Hence there is 85% chance that the house will be sold

P = Prob. of Selling of the car

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$\hat{y} = 0.231 + 0.5(x_1) + 0.9(x_2)$$

$$z = 5.677$$

In log-reg case the car will be sold as $P \geq 0.5$

$P \geq 0.5 \rightarrow$ YES, 1, SELL

$P < 0.5 \rightarrow$ NO, 0, NOT SELL

SIGMOID FUNCTION:

OR
logit functi.

$$f(z) = \sigma_z = \frac{1}{1 + e^{-z}}$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$f(z) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

Exp: $z = 5.667$ (As from prev. example)

$$f(5.667) = \frac{1}{1 + e^{-5.667}} = 0.996 \rightarrow \text{SELL THE HOUSE}$$

LOGISTIC REGRESSION SETUP AND WORKING:

Exp: 1

(Study hrs) x	(GPA(out of 10)) y	(expected GPA) $\hat{y} = \beta_0 + \beta_1 x$	(Prob. of pass) $f(z) = \frac{1}{1 + e^{-z}}$	STATUS
2	6	3.5099	$\frac{1}{1 + e^{-3.5099}} = 0.97$	YES
3	2	4.2566	0.98	YES
4	9.8	8.7368	0.99	YES
6	5.2	6.4967	0.99	YES

β_0, β_1 to be solved using SLR method.

$$\text{Hence } \hat{y} = 2.0165 + 0.7467x$$

$$\text{Then } \ln\left(\frac{p}{1-p}\right) = 2.0165 + 0.7467x$$

$$\text{for } x=2 \quad \ln\left(\frac{p}{1-p}\right) = 2.0165 + 0.7467(2) = 3.5099$$

(TASK) Exp: 2

A smart hospital uses AI to predict length of stays and ICU admission based on BMI, Blood Pressure and Oxygen Saturation

Sample data for practice:

BMI	BP	OS	y_1	y_2
31	150	88	12	1
24	120	97	4	0
29	140	92	8	1

- model the data
- Predict length of stay when BMI is 25, BP is 160 and OS is 90?
- Also find the prob. of stay for above patient
- Decide he will have to stay or not?